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Review

Interdigitated array microelectrodes based impedance biosensors for detection of bacterial cells

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ABSTRACT

Impedance spectroscopy is a sensitive technique to characterize the chemical and physical properties of solid, liquid, and gas phase materials. In recent years this technique has gained widespread use in developing biosensors for monitoring the catalyzed reaction of enzymes; the bio-molecular recognition events of specific proteins, nucleic acids, whole cells, antibodies or antibody-related substances; growth of bacterial cells; or the presence of bacterial cells in the aqueous medium. Interdigitated array microelectrodes (IDAM) have been integrated with impedance detection in order to miniaturize the conventional electrodes, enhance the sensitivity, and use the flexibility of electrode fabrication to suit the conventional electrochemical cell format or microfluidic devices for variety of applications in chemistry and life sciences. This article limits its discussion to IDAM based impedance biosensors for their applications in the detection of bacterial cells. It elaborates on different IDAM geometries their fabrication materials and design parameters, and types of detection techniques. Additionally, the shortcomings of the current techniques and some upcoming trends in this area are also mentioned.

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1. Introduction

Electrochemical biosensors, often referred to as amperometric, potentiometric, conductimetric, or impedimetric, are advantageous as they are highly sensitive, rapid, inexpensive, and are suitable for designing integrated microsystems (Radke and Alocilja, 2005a; Bakkar and Qin, 2006). Electrochemical impedance combines the analysis of the resistive and capacitive (or inductive) properties of materials in response to the small amplitude sinusoidal excitation signal (Bott, 2001; Guam et al., 2004). Impedance detection works by measuring the impedance change caused by binding of target molecules to receptors (antibodies, DNA, proteins, and other bio-recognition elements) immobilized on the surface of the electrodes (Yang et al., 2004a; Radke and Alocilja, 2004, 2005a,b;

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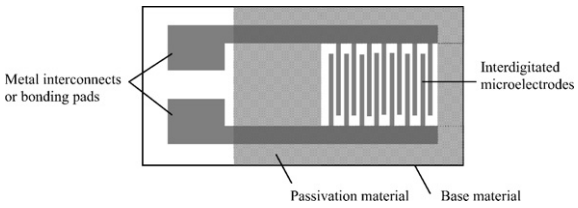


Fig. 1. The schematic design of a typical planar interdigitated array microelectrode chip. Metal interconnects or bonding pads are for connecting electrodes to the impedance analyzer and the interdigitated microelectrodes are for sensing. The area around electrodes is passivated to collect the electrical response of the interdigitated microelectrodes only.

Yang and Li, 2005), change in the conductivity of the medium caused by the growth of bacteria (Gomez et al., 2001, 2002; Gomez-Sjöberg et al., 2005; Yang et al., 2004b; Yang and Li, 2006), change in conductivity of the medium due to suspension of target molecule in the aqueous medium (Varshney and Li, 2007a; Varshney et al., 2007b), capturing bacterial cells on the surface of electrodes using dielectrophoresis (DEP) (Li and Bashir, 2002; Suehiro et al., 1999, 2001, 2003a,b; Aldaeus et al., 2005), and change in the ionic concentration of the medium caused by the activity of enzyme used as labels for the signal amplification (Laureyn et al., 2000; Ruan et al., 2002; Kim et al., 2004; Thomas et al., 2004). Generally impedance measurement is divided into two categories: faradic and non-faradic (Yang et al., 2004a). Faradic requires a redox probe for impedance measurement, while non-faradic measurement can be performed in the absence of a redox probe. Traditionally, macro-sized metal rods or wires were used as electrodes immersed in the medium to measure impedance (Towe and Pizziconi, 1997; Berggren et al., 1998; Mirsky et al., 1998). In an attempt to miniaturize the sensor and improve the sensitivity, microelectrodes have been considered as potential candidate to combine with traditional detection systems. Microelectrodes favor a greater rate of reactant supply (while macroelectrodes cause greater depletion of reactants) and require lower concentrations of electro-active ions to form double layer as compared to macroelectrodes (Ciszkowska and Stojek, 1999; Min and Baeumner, 2004). As a result, microelectrodes can perform impedance measurement even in low conductivity solution, where macroelectrodes may not be sensitive.

Among microelectrodes, interdigitated array microelectrodes (IDAM) present promising advantages in terms of low ohmic drop, fast establishment of steady-state, rapid reaction kinetics, and increased signal-to-noise ratio (Amatore et al., 1983; Ciszkowska and Stojek, 1999; Mauyama et al., 2006). IDAM consist of a series of parallel microband electrodes in which alternating microbands are connected together, forming a set of interdigitating electrode fingers (Fig. 1). Due to proximity of cathodic and anodic electrodes, minute amounts of ionic species can be efficiently cycled between the electrodes resulting in very large (>0.98) collection efficiencies, giving the IDAM an advantage in detecting small amounts of generated electrode products (Postlethwaite et al., 1996; Thomas et al., 2004). Additionally, IDAM eliminates the need for a reference electrode and provides simple means for obtaining a steady-state current response, which is comparatively simpler to detect as compared to three or four electrode systems (Liu et al., 2004; Nebling et al., 2004). Their low response time also favors rapid detection. The typical dimensions of an individual microband "finger" are 0.1–0.2 μm in height, 1–20 μm in width; 2–10 mm in length, with a gap of 1–20 μm between the fingers.

2. Electrode material

The electrode material is critical for the sensitivity and selectivity of the impedance system. IDAM used for impedance detection are made of gold (Au), platinum (Pt), titanium (Ti), chromium (Cr), carbon (C), and indium tin oxide (ITO) (Iwasaki and Morita, 1995; Liu et al., 2004; Hayashi et al., 2005; Toriello et al., 2005; Kim et al., 2006). They are fabricated on a variety of base materials, but the most commonly used ones are silicon, quartz/glass, and aluminum oxide (Nebling et al., 2004; Toriello et al., 2005; Kim et al., 2006). Different shapes of the electrodes are used such as planar interdigitated electrodes, octagonal interdigitated ring electrodes, multi-islands of interdigitated electrodes, ring array electrodes, and 3-D micromesh electrodes (Niwa and Morita, 1996; Liu et al., 2000; Nebling et al., 2004; Hayashi et al., 2005; Sato et al., 2007). To avoid complexity of the fabrication process, only one material is typically used for electrode fabrication, however some work has studied the effect of using different combinations of materials.

The selection of materials for electrode fabrication depends on the intended application, ionic species involved, inertness of the material to the environment, and their suitability to the fabrication process. For example, ITO is used in optoelectronic devices because of its high transparency and low resistance. It is used as an oxidation electrode and is selective against anionic species. To obtain a high sensitivity for electrochemical detection, it can be combined with a highly reactive metal like gold to reduce oxidized products and provide continuous redox cycling (Hayashi et al., 2005). Other material such as carbon has wider potential window and larger over-potential for water as compared to noble metal electrodes. This feature is expected to improve the sensitivity of electrochemical reactions for redox-active biological molecules because their reactions are slow on metal electrodes (Iwasaki and Morita, 1995).

3. Parameters in electrode design

Various design parameters have been evaluated to understand their effects on the sensitivity and overall working of the IDAM such as the number, width, height and length of electrodes as well as the gap (or spacing) between electrodes. Initially, it was believed that only the ratio of the gap and width of the electrodes was a key to improving the sensitivity of IDAM, but Min and Baeumner (2004) showed that the height and material of electrodes are also important contributing factors. It was shown that there is no significant effect of number of fingers on signal to noise (S/N) ratio as compared to other parameters, because the signal value is proportional to the surface area of the whole array and the background noise is proportional to the area of the electrodes only (Stulik et al., 2000), therefore, increase in the number of fingers will not only increases the signal value but also the background noise, resulting no change in S/N ratio. The S/N ratio was increased with decrease in the electrode width, owing to the combined spherical and vertical diffusion field of the electrodes with smaller electrode width (Stulik et al., 2000).

The performance of IDAM was quantitatively analyzed for a wide range of electrode spacing (Bard et al., 1986; Fosset and Amatore, 1991, Jin et al., 1996; Cohen and Kunz, 2000; Stulik et al., 2000). In devices with spacing as low as 800 nm, the redox cycling efficiencies were of the order of 40. Proximity of electrodes resulted in an increased sensitivity and a short response time because of the enhanced redox cycling efficiency (i.e., faradic current enhancement), fast establishment of steady-state, rapid kinetics of reaction, and reduced amount of ions diffusing back to the bulk (Aoki, 1988; Iwasaki and Morita, 1995; Cohen and Kunz, 2000). The height of the electrodes was directly correlated with overall signal and S/N ratio. The S/N ratio reached a peak value at certain value of electrode height and then took a downward trend (Min and Baeumner, 2004). The increased signal value and S/N ratio were the result of availability of more surface area available due to increase in the height of electrodes.

4. Types of impedance detection techniques

Impedance techniques used for bacterial cell detection are broadly categorized on the basis of the use of bio-recognition element on the surface of the electrodes, non-specific adsorption of bacterial cells on the surface of electrodes, detection of metabolites produced as a result of the bacterial growth, and detection based on the charge of a bacterial cell or its internal components. Different categories of impedance detection techniques along with their applications are discussed in the following sections.

4.1. Detection based on the use of specific bio-recognition element on the surface of electrodes

In order to study particular target analytes, such as bacterial cells or nucleic acid extracted from bacterial cells, electrode surfaces have been functionalizedwith bio-recognition elements (generally antibodies or nucleic acid). A typical format for these sensors is shown in Fig. 2a. Different surface modification techniques are used for the immobilization of bio-recognition elements on the surface of electrodes and are selected based on the electrode material and the addressable functional groups present (or modified) on the surface of bio-recognition elements and the surface of the electrodes.

The surface of the electrode can be conjugated with and without the use of chemical linker. Yang et al. (2004a) used an open ITO IDAM chip coated with anti-*E. coli* antibodies by spreading antibodies on the chip and storing it overnight at 4 °C. The antibodies and captured *E. coli* O157:H7 cells behaved as insulators and increased the electron-transfer resistance measured directly in the presence of ferri/ferrocyanide. The detection range of the biosensor was 4.36×10^5 to 4.36×10^8 CFU ml^{-1}. Radke and Alocilja (2004) used an open gold IDAM chip immobilized with antibodies by silanizing the sensor surface using 3-mercaptomethyldimethylethoxysilane and a heterobifunctional cross-linker, N-(g-Maleimidobutyryloxy) succinimide ester. The optimum width and spacing for the electrode fingers were determined to be 3 and 4 μm, respectively, based on the presence of 90% of total electric field strength below a distance of 5 μm from the sensor surface. Impedance measurement was performed by immersing electrode arrays into samples of *E. coli* O157:H7 cells suspended in peptone water. The biosensor was found to be sensitive for the detection of *E. coli* O157:H7 ranging from 10^5 to 10^7 CFU ml^{-1}. Using the same methodology and electrode setup, they were able to improve the minimum detection limit from 10^5 to 10^4 CFU ml^{-1} in pure cultures and romaine lettuce wash water. The sensor surface was selective for the detection of target bacteria, *E. coli* O157:H7 in presence of non-target bacteria, *Salmonella infantis* (Radke and Alocilja, 2005a,b).

A limited work has been done on the use of nucleic acids as bio-recognition elements for the electrode functionalization. Elsholz et al. (2006) developed a 16S RNA based, electrical oligonucleotide microarray for identifying and quantifying five different pathogens (*E. coli*, *P. aeruginosa*, *Enterrococcus faecalis*, *Staphylococcus aureus*, and *Staphylococcus epidermidis*). Thiol-modified oligonucleotides probe were immobilized on the gold IDAM, mediating the specific detection of the target 16S rRNA by hybridization. Biotin-labeled oligonucleotides were hybridized with the probe sequence and were subsequently combined with avidin-alkaline phosphatase conjugates for the generation of an electrical signal after the addition of a substrate. The signal changed over the time proportional to the analyte concentration. The detection limit for *E. coli* total RNA was 0.5 ng μl^{-1}. The total detection time with this fully automated system was less than 25 min.

The use of bio-recognition element directly on the surface of electrodes is the most conventional for impedance detection and could be considered advantageous in terms of directly providing specificity to the biosensor. The antigen–receptor reactions on the surface of the electrodes bring antigens within few nanometers of the sensing electrodes, maximizing the effect each antigen has on impedance.

4.2. Detection based on the non-specific adsorption (without the use of bio-recognition element) of bacterial cells on the surface of electrodes

In this category, no bio-recognition elements are used on the surface of electrodes for the capture of bacterial cells, rather magnetic beads coated with bio-recognition element are used for the specific separation and concentration of cells prior to impedance measurement. The schematic of the impedance technique using immunomagnetic beads is shown in Fig. 2b. Previous research has pointed out that when antibodies were immobilized on the surface of electrodes, the functional surface area (where target bacterial cells are detected) of the electrode is not optimally utilized. The experimental data showed that anti-*E. coli* antibodies immobilized on the surface of ITO coated glass electrodes displayed only 16% capture efficiency (CE) for *E. coli* O157:H7 (Ruan et al., 2002) and the CE of anti-*Salmonella* antibodies functionalized on a roughened glass surface was less than 1% for *Salmonella* (Brewster et al., 1996).

Varshney and Li (2007a) developed a label-free impedance biosensor based on gold IDAM for the detection of *E. coli* O157:H7 in food samples. *E. coli* O157:H7 cells were separated and concentrated with the help of magnetic nanoparticle-antibody conjugates (MNAC), functionalized with anti-*E. coli* antibodies. After separation and resuspension of bacterial cells in low conductivity 0.1 M mannitol solution, the concentrated sample was uniformly spread on the surface of IDAM. The impedance sensor detected a minimum of 7.4×10^4 and 8.0×10^5 CFU ml^{-1} of *E. coli* O157:H7 in pure culture and ground beef samples, respectively. The concentration of bacterial cells attached to MNAC in the active layer of IDAM above the surface of electrodes was achieved with the help of a magnet placed underneath the electrode chip. This improved the sensitivity of the biosensor by 35% as compared to no use of the magnet. These experiments were performed on an open chip. The same technique was used to detect *E. coli* O157:H7 in the flow cell embedded with gold IDAM (Varshney et al., 2007b). The flow cell was used to detect bacterial cells in a detection volume of 60 nl. This impedance biosensor was able to detect as low as 1.6×10^2 and 1.2×10^3 cells of *E. coli* O157:H7 cells present in pure culture and ground beef samples, respectively. The total detection time from sampling to measurement was 35 min. In both of these experiments, immunomangnetic separation (IMS) based on nanoparticles was used to separate target bacterial cell from the food matrix and avoid the tedious processes of filtration and centrifugation for sample preparation (Varshney et al., 2005).

This technique has a major advantage of additional separation and concentration of bacterial cells from crude samples before impedance measurement. In this way, the background noise caused by the non-target components in the samples could be reduced, improving the signal to noise ratio. Additionally, the surface of electrodes can be used multiple times as there is no bio-recognition element present directly on the surface of electrodes.

4.3. Detection based on metabolites produced by bacterial cells as a result of growth

The detection of metabolites produced as a result of bacterial growth is also known as impedance microbiology. This technique is applied in a variety of fields ranging from detection and monitoring of microorganisms, detection of antibiotics, analysis of food preservatives, food hygiene, clinical and pharmaceutical

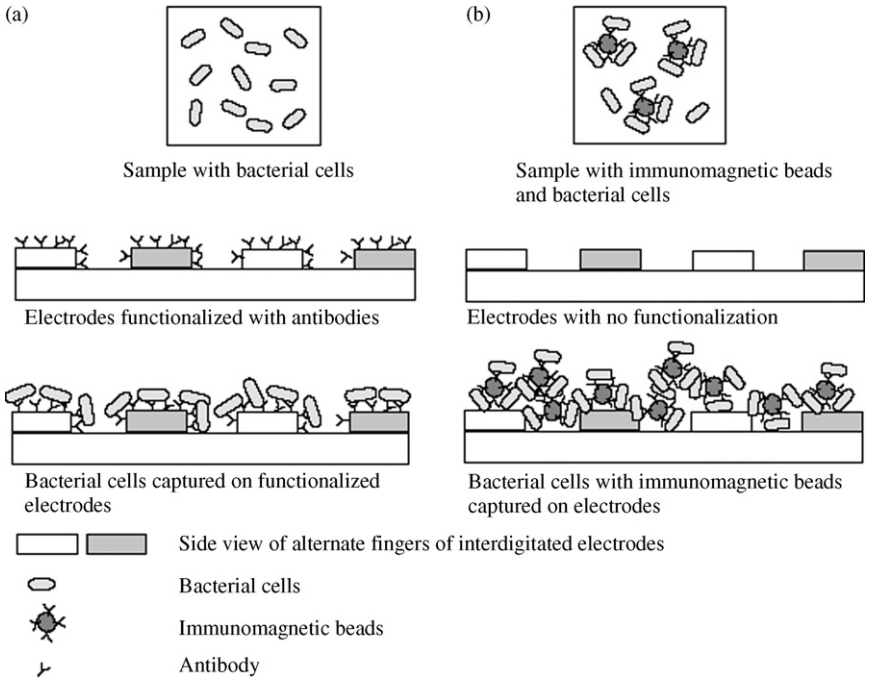


Fig. 2. Working principle of impedance detection technique (A) based on the use of bio-recognition elements, and (B) based on the non-specific adsorption (without the use of bio-recognition element) of bacterial cells on the surface of electrodes.

microbiology to environmental sampling (Okigbo and Richardson, 1985; Strassburger et al., 1991; Kroyer et al., 1995; Silley and Forsythe, 1996; Wawerla et al., 1999; Gomez-Sjöberg et al., 2005). This indirect approach to quantitative microbiology enumerates microorganisms by measuring the change in the electrical conductivity of the medium during growth of microorganism (Strassburger et al., 1991). The growth of microorganisms increases the conductivity of the medium by converting uncharged or weakly charged substances present in the growth medium, such as yeast, peptone, and sugar into highly charged substances such as amino acids, aldehydes, ketones, acids, and other metabolic products (Wawerla et al., 1999). A temperature controlled growth chamber with suspended IDAM is used for the measurement of the growth of bacterial cells (Fig. 3). Since only live bacterial cells can cause change in the conductivity of the medium (due to metabolic by-products produced), impedance microbiology is also used for differentiating live and dead bacterial cells (Felice and Valentinuzzi, 1999; Gomez et al., 2002; Suehiro et al., 2003a). However, it is still challenging to differentiate between injured bacterial cells and metabolically active bacterial cells.

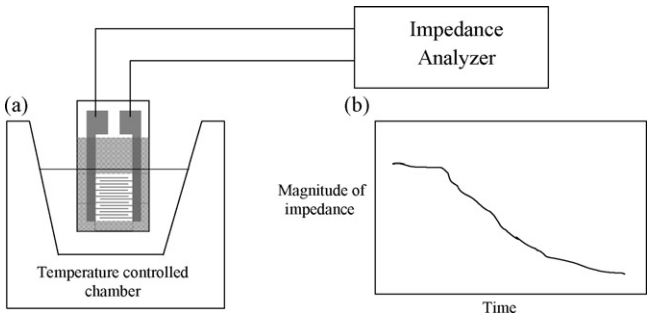


Fig. 3. (a) Experimental set-up of the indirect impedance measurement with interdigitated array microelectrodes for the detection of bacterial growth. (b) The graph shows the change in the magnitude of impedance with time for the growth of bacteria.

The sensitivity of impedance detection depends on design of the growth medium in addition to other parameters previously discussed. The ideal medium enhances the growth of the bacterial cells without a significant contribution of the conductivity of its substrates to the overall conductivity of the medium. Gomez et al. (2002) developed a flow cell embedded with platinum IDAM to detect metabolic activity of few live bacterial cells. A low conductivity medium was specially designed to maximize the change in conductance of the medium due to the growth of *L. innocua*, *E. coli*, and *L. monocytogenes*. In the same experimental setup, live and heat-killed bacterial cells were also differentiated. After 2 h of off-chip incubation, the minimum number of live cells suspended in a low conductivity buffer that could easily be distinguished from the same number of heat-killed cells was in the order of 100 *L. innocua*, 200 *L. monocytogenes*, and 40 *E. coli* cells, confined into a volume of 5.27 nl.

In a biosensor, the specificity of the system is achieved with the help of bio-recognition elements. However, in impedance microbiology either selective medium or microbial growth in a non-selective medium following IMS is used to impart specificity. Yang et al. (2004b) used open ITO IDAM for the detection of viable *Salmonella typhimurium* in milk samples using a selective medium. Selenite cystein broth supplemented with trimethylamine oxide and mannitol was used for the selective growth of *S. typhimurium*. Three other non-target common foodborne pathogenic bacteria, including *Listeria monocytogenes*, *E. coli* O157:H7, and *Pseudomonas aeruginosa* were used for testing the selectivity of the medium for *S. typhimurium*. The impedance system was successfully used for the detection of a range of *S. typhimurium* from 4.8 to 5.4×10^5 CFU ml^{-1} after an enrichment growth of 9.3 and 2.2 h, respectively. The formulation of selective medium with low conductivity is a challenging task; therefore, the use of a non-selective media in conjunction with IMS was introduced by Yang and Li (2006). Anti-*Salmonella* antibodies coated immunomagnetic beads were used to separate *S. typhimurium* from the samples and was followed by the growth of bacterial cells in a non-selective brain heart infusion broth. It was observed that the impedance system was able to detect

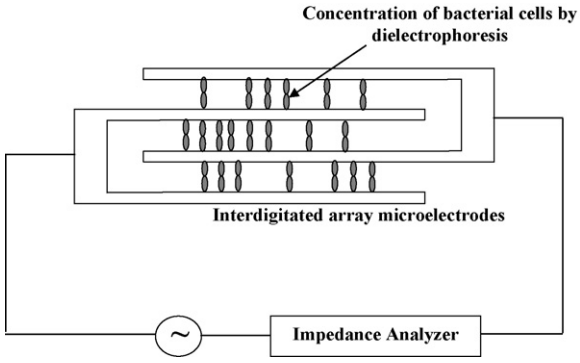


Fig. 4. Dielectrophoresis based impedance detection system (adopted from Suehiro et al., 2003b) for the simultaneous concentration and impedance detection of bacterial cells.

initial concentrations of 10^1 and 10^6 CFU ml^{-1} of *S. typhimurium* in a growth time of 8 and 1.5 h, respectively.

Impedance microbiology is combined with other techniques in the design of a fully automated IDAM based impedance biochip. Gomez-Sjöberg et al. (2005) combined dielectrophoresis (DEP) with impedance microbiology for the design of an on-site incubation microfluidic biochip device embedded with platinum IDAM for the detection of *Listeria*. Concentration and capture of the cells inside the biodevice was achieved by applying DEP force. DEP was successfully used for the concentration of bacterial cells by a factor of 10^4 to 10^5 in a detection chamber of 400 pl. The detection time for 8.0×10^4 CFU ml^{-1} of *L. monocytogenes* was less than 2 h.

Impedance microbiology is beneficial for real-time monitoring of biological activity of bacterial cells, their metabolites, and other related kinetics which are not possible with any of the other impedance technique. This technique is also beneficial in discriminating between live and dead bacterial cells that could be important to detect the contamination of food by toxins produced by bacterial cells. Additionally, the surface of electrodes can be reused for several different tests as there is no bio-recognition element used on the surface of electrodes.

4.4. Detection based on the charge of a bacterial cell or its internal components

Since bacterial cells and its components are charged, it can be polarized in the presence of an electric field using DEP force. Microelectrodes use small operating voltages to generate high electric field strength, thus allowing them to achieve high DEP force without generation of excessive electrical heating and related electrochemical effects (Pethig and Markx, 1997). Although any arbitrary geometry of planar electrodes can be easily designed and fabricated with modern lithography, the three most commonly used geometries are: simple gap, castellated, and quadrupole electrodes (Aldaeus et al., 2005; Menachery and Pethig, 2005). Here, in this paper we are limiting our discussion to interdigitated electrodes only. Interdigitated electrodes are used to generate positive dielectrophoresis, where particles are attracted to the edges of the electrodes; or negative dielectrophoresis, where particles are pushed away from the plane of the electrodes. A schematic of the DEP based impedance detection system is shown for the concentration and detection of bacterial cells (Fig. 4).

DEP force is used for aligning bacterial cells between the electrodes and is followed by the conventional impedance measurement. Suehiro et al. (1999) used dielectrophoretic impedance measurement (DEPIM) for the concentration of the bacterial cells in the form of "pearl chain" between the electrodes followed by an impedance measurement. Under flow condition, the detection system was able to detect 10^5 CFU ml^{-1} in 10 min. In the same system, live and dead bacteria were differentiated by their response to the frequency of the applied voltage (Suehiro et al., 2003a). In an approach to improve the detection limits of the DEP based impedance biosensors, the intracellular ions of the cell are expelled in the environment by electropermeation (EP) using bipolar rectangular pulses in place of a sinusoidal ones. Bipolar rectangular pulses are reported to have higher EP efficiency than sinusoidal pulses of the same frequency as pulse amplitude for the former exceeds a critical value for EP for a longer duration (Kotnik et al., 2001a,b, 2003). Suehiro et al. (2003b) designed an electropermeabilization-assisted DEPIM (EPA-DEPIM) to improve the sensitivity of the conventional DEPIM. The improved method decreased the detection limit of DEPIM for yeast cells from 10^4 to 10^2 CFU ml^{-1} in 15 min diagnostic time. The improved system was also used for the detection of 10^2 CFU ml^{-1} of *E. coli* in a detection time of 3 h (Suehiro et al., 2005).

Although DEP based systems do not fall into the category of a biosensor as they lack bio-sensing element responsible for the specificity of the system, nevertheless, Suehiro's group has presented some work to show that the specificity can also be added to DEP devices. Suehiro et al. (2001) used a IDAM embedded microfluidic flow cell coupled with antigen–antibody reaction for concentration and specific impedance detection of *E. coli* cells. This study was presented as a proof of concept and only a high concentration (10^8 CFU ml^{-1}) of *E. coli* cells was used. In another study by Suehiro et al. (2006), anti-*E. coli* antibodies were immobilized on the surface of the IDAM chip and impedance measurement was performed in the presence of low and high DEP forces. The effect of EPA-DPIM was also observed with low DEP force.

The application of DEP force to concentrate bacterial cells has shown to expose some proteins on the surface of bacterial cells that has resulted in an increased capture of bacterial cells by antibodies. Yang et al. (2006) developed an IDAM based multifunctional microfluidic system for the concentration and specific capture of *L. monocytogenes* by employing DEP and immobilizing anti-*Listeria* antibodies, respectively. Bacterial cells were attracted to the edges of the electrodes because of positive DEP, resulting in a concentration of bacterial cells by a factor of 10^2–10^3 in a sample volume of 5–20 μl. Cells concentrated by DEP were captured by antibodies immobilized on the channel surface with the efficiencies of 18–27% when bacterial cell numbers ranged from 10^1 to 10^3 cells. Scanning electron micrographs clearly showed that more proteins were exposed on the surface of the bacterial cells after DEP, causing increased number of binding sites for antibodies. They were also able to differentiate between live and dead bacterial cells simply by monitoring the response of bacterial cells to the frequency used in DEP. In another work, an impedance based method was used to detect viable spores of *Bacillus anthracis* by electrically detecting their germination in real time within a microfluidic chip. A three layer PDMS microfluidic biochip with valves, flow channel, and interdigitated electrodes was used for the concentration of spores by DEP and germination, followed by the impedance measurement. The detection limit of the biochip was 10^9 spores ml^{-1} (Liu et al., 2007).

The DEP based impedance detection requires almost no sample preparation step, is rapid and simple to use, however, the discrimination of target bacterial cells from non-targeted one and selection of non-interfering medium are challenging, as discussed later in Section 6.

Table 1 shows the list of all IDAM based impedance techniques, medium used for measurement and their relative conductivity with respect to the conductivity of the bacterial cells, bio-recognition element used, and various applications for bacterial detection.

Table 1
Comprehensive list of working principles.

Working principle	Medium[a]	Relative conductivity[b]	Bio-affinity element	Microorganism	System	Typical application	Reference
Use of bio-recognition element	Redox probe	High	Polyclonal antibodies	*E. coli* O157:H7	IDAM chip	Detection of bacterial cells. Detection limit: 10^6 CFU ml^{-1}	Yang et al. (2004a)
	Peptone water	High	Polyclonal antibodies	*E. coli* O157:H7	IDAM chip	Detection of bacterial cells in pure cultures. Detection limit: 10^4, 10^5 CFU ml^{-1}	Radke and Alocilja (2004, 2005a)
	Peptone water	High	Polyclonal antibodies	*E. coli* O157:H7	IDAM chip	Detection of bacterial cells in food samples. Detection limit: 10^4 CFU ml^{-1}	Radke and Alocilja (2004, 2005a,b)
	Mannitol solution	Low	Magnetic nanoparticles-antibody conjugates	*E. coli* O157:H7	IDAM chip	Detection of bacterial cells in food samples. Detection limit: 7.4×10^4 CFU ml^{-1}	Varshney and Li (2007a)
	Mannitol solution	Low	Magnetic nanoparticles-antibody conjugates	*E. coli* O157:H7	Flow cell with embedded IDAM	Detection of bacterial cells in food samples. Detection limit: 1.6×10^2 cells	Varshney et al. (2007b)
Impedance microbiology	Growth medium	High	X[c]	*S. typhimurium*	IDAM chip	Detection of bacterial growth. Detection limit: 4.8×10^0 CFU ml^{-1}	Yang et al. (2004b)
	Growth medium	Low	–	*L. innocua, E. coli, L. monocytogenes*	Flow cell with embedded IDAM	Detection of bacterial growth and differentiation of live and dead cells. Differentiation of 40–200 bacterial cells of different strains	Gomez et al. (2002)
	Growth medium	Low	–	*L. innocua*	Flow cell with embedded IDAM	Detection of bacterial growth. Detection limit: 1 bacterium	Gomez et al. (2001)
Impedance microbiology and immunomagnetic separation	Growth medium	High	Antibodies conjugated magnetic microbeads	*S. typhimurium*	IDAM chip	Detection of bacterial growth. Detection limit: 10^1 CFU ml^{-1}	Yang et al. (2006)
Dielectrophoresis and impedance detection	Mannitol solution	Low	X[d]	*E. coli*	Flow cell with embedded IDAM	Concentration and detection of bacteria. Detection limit: 10^5 CFU ml^{-1}	Suehiro et al. (1999)
	Mannitol solution	Low	–	*E. coli*	Flow cell with embedded IDAM	Differentiation of live and dead bacterial cells. No detection limit: only proof of concept	Suehiro et al. (2003a)
	Mannitol solution	Low	Polyclonal antibodies	*E. coli* O157:H7	Flow cell with embedded IDAM	Concentration and detection of bacteria. No detection limit: just proof of concept	Suehiro et al. (2001)
Dielectrophoresis, immunoaffinity, and impedance measurement	Mannitol solution	Low	Polyclonal antibodies	*E. coli*	Flow cell with embedded IDAM	Concentration and detection of bacteria. Detection limit: 10^6 CFU ml^{-1}	Suehiro et al. (2006)
Dielectrophoresis, electropermeation and impedance detection	Mannitol solution	Low	X[d]	*E. coli*	Flow cell with embedded IDAM	Concentration and detection of bacteria. Detection limit: 10^2 CFU ml^{-1}	Suehiro et al. (2003b, 2005)
Nucleic acid based impedance detection	–	–	Oligonucleot-ides	*E. coli, P. aeruginosa, Enterrococcus faecalis, Staphylococcus aureus, and Staphylococcus epidermidis*	IDAM chip	Nucleic acid detection of bacterial cells by enzymatic amplification using alkaline phosphatase. Detection limit: 0.5 ng μL^{-1} of total RNA	Elsholz et al. (2006)

[a] Medium (in which impedance was measured).
[b] Relative conductivity (relative to the conductivity of bacterial cells), bio-affinity element, system, and applications of interdigitated array microelectrodes based impedance biosensors for detection of bacterial cells.
[c] There was no bio-recognition element used. Selective growth medium were used to grow bacterial cells to cause change in the conductivity of the medium.
[d] Mannitol was used as a low conductivity medium for measuring the impedance change due to the presence of bacterial cells.

5. Equivalent circuit analysis

In an electrochemical assay, the medium resistance and other electrical parameters are not directly calculated from the input voltage, current, and electrode area. Instead, an equivalent circuit is used to curve fit the experimental data and extract the necessary information about the electrical parameters responsible for the impedance change. Direct calculations are not possible due to non-uniform current distribution through an electrolyte, and the difficulty in determining the current flow path. Thus an equivalent circuit representing the experimental data is helpful in determining the most sensitive element(s) responsible for the impedance change due to the presence of bacterial cells.

The electrical parameters generally used to design an equivalent circuit model for electrochemical curve fitting include electrolyte resistance (bulk medium resistance), double layer capacitance, dielectric capacitor, polarization resistance, charge transfer resistance, Warburg impedance, interfacial impedance, coating capacitance, and virtual inductors. The types of electrical components in the model and their interconnections control the shape of the equivalent circuit's impedance spectrum, and the circuit parameters control the size of each feature in the spectrum. Both of these factors affect the degree to which an equivalent circuit model represents the experimental impedance data. While more than one circuit model can fit the experimental data, a unique equivalent circuit is selected based on the circuit parameters best explaining the electrochemical cell's physical characteristics. Yang et al. (2004b) designed an equivalent circuit comprising a double layer capacitor, dielectric capacitor, and medium resistor to explain the behavior of an IDAM based impedance sensor for the detection of viable *S. typhimurium* after enrichment growth in a medium. The circuit parameters dominated the impedance change in different ranges of frequencies. The double layer capacitance, medium resistance, and dielectric capacitance dominated the total impedance from 0.2 Hz to 50 Hz, 1 kHz to 1 MHz, and >1 MHz, respectively. In an another study related to the growth of the bacterial cells, Gomez-Sjöberg et al. (2005) used an equivalent circuit model consisting of one capacitor (for dielectric capacitance), four resistors (one for bulk resistance, one for interfacial resistance for each pair of electrodes, and one for on-chip wiring) to curve fit the experimental data. Two interfacial resistors for each electrode were used to account for the interaction of the electrode with two distinct groups of species in the solution. A dielectric capacitor was added to the circuit to account for the dielectric properties of all the materials surrounding the electrodes. The effect of ionic species in the solution was represented by the bulk resistance of the liquid.

The circuit model for the IDAM impedance system based on the use of antibodies immobilized on the surface of electrodes for the capture of target bacteria consisted of double layer capacitance, dielectric capacitance and solution resistance (Radke and Alocilja, 2005a). It was suggested that double layer capacitance and solution resistance contributed to the total impedance at lower frequencies (<1 MHz), while dielectric capacitance contributed to the total impedance at higher frequencies (>1 MHz). Impedance measurement was performed in the presence of peptone water. Impedance systems based on highly conductive redox probes have also been modeled by an equivalent circuit. Yang et al. (2004a) used an open ITO IDAM chip immobilized with anti-*E. coli* antibodies for capture of *E. coli* O157:H7 cells. The impedance response of the system was measured in the presence of redox probe, ferri/ferrocyanide. The equivalent circuit consisted of an ohmic resistor of the electrolyte between two electrodes, double layer capacitor, an electron-transfer resistor, and Warburg impedance around each electrode. Experimental data fitting to the equivalent circuit showed that the electron transfer resistance and electrolyte resistance were responsible for the detection of *E. coli* O157:H7 cells.

Several experiments have been conducted without any surface modification of the electrodes for direct impedance detection of bacterial cells, but very few have presented the equivalent circuit model. Varshney and Li (2007a) showed with the help of equivalent circuit analysis that bulk resistance and double layer capacitance were responsible for the impedance change caused by the presence of *E. coli* O157:H7 on the surface of IDAM. In another study conducted by the same group, an equivalent circuit consisting of bulk resistor, double layer capacitor and dielectric capacitor was used to explain the behavior of *E. coli* O157:H7 with MNAC inside microfluidic flow cell with embedded IDAM (Varshney et al., 2007b).

A comprehensive list of equivalent circuit models used for representing experimental impedance data of IDAM based detection systems for the detection of bacterial cells is presented in Table 2 .

6. Shortcomings of the present research

Impedance techniques based on IDAM have been used in a number of applications and researchers have endeavored to constantly improve the sensitivity, functionality, and detection limit of this technique. However, it has some shortcomings that need to be addressed in order to further advance this technique. These shortcomings are not necessarily inherent to the impedance detection system but can also come from the limitations of the individual techniques combined with the impedance detection.

Both faradic and non-faradic impedance measurement can be performed by capturing bacterial cells on the surface of electrodes by the antibodies immobilized on IDAM chip. Although this has been used in numerous applications, a major problem associated with antibody immobilization is the low CE of the immobilized surface. This issue needs to be addressed by either developing better surface chemistry for the immobilization of a dense and uniform layer of antibodies or by developing impedance methods without immobilization of bio-affinity material on the electrodes. Separation of target bacterial cells using immunomagnetic particles has successfully replaced separation of bacterial cells by immobilizing antibodies on the surface of electrodes. Immunomagnetic particles are efficient in separating target bacterial cells from complex background and show a range of CE from 60% to 100% depending on the immunoreaction time, food matrix, and concentration of target bacterial cells in the sample (Varshney and Li, 2007a; Yang and Li, 2006). Despite their excellent capability to capture bacterial cells, they have some disadvantages such as interference from the impedance of the particles alone and cluster formation with bacterial cells. Cluster formation is more common with magnetic nanoparticles (Varshney et al., 2005; Varshney and Li, 2007a) and prevents formation of a uniform layer of bacterial cells on the surface of electrodes. These problems require attention to improve the performance of techniques based on the integration of IMS with impedance detection.

Impedance microbiology is quite mature for the impedance detection of microbial growth. However, better design of a growth medium in terms of low conductivity and rapid growth of bacterial cells will be required for the rapid and sensitive impedance detection of bacterial cells.

To detect bacterial cells directly from food samples, interference from food matrix needs to be reduced. The food/sample matrix and other non-targeted bacterial cells interfere with the impedance detection of target bacterial cells as they are similar to target cells in terms of surface charge and change in the conductivity of the solution. Although, there are several separation techniques available such as, IMS, DEP, surface functionalization of electrodes, and differential growth medium. However, improving current techniques

Table 2
Equivalent circuit models used in interdigitated array microelectrodes based impedance detection systems for the detection of bacterial cells.

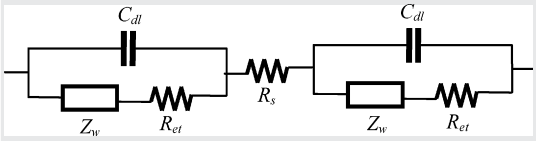
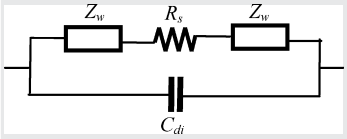
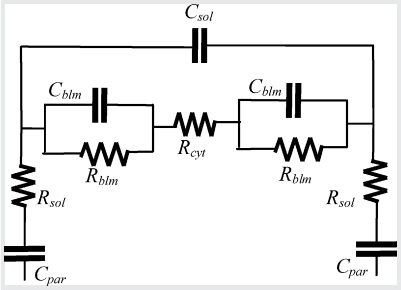
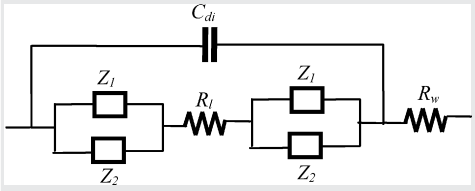
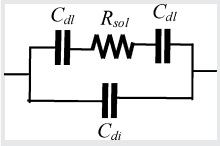
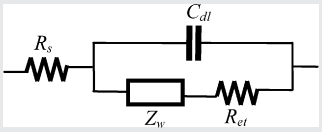
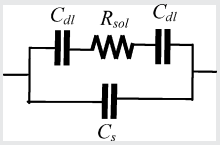
Equivalent circuits	Typical applications	Electrical circuit elements	References
	Label-free electrochemical immunosensor for detection *of E. coli* O157:H7 by immobilizing anti-*E. coli* antibodies on the surface of indium-tin oxide IDAM and measuring impedance change in the presence of a redox probe	C_{dl} is double layer capacitance R_{et} is electron-transfer resistance Z_w is Warburg impedance R_s is the resistance of electrolyte solution	Yang et al. (2004a)
	Label-free detection of Salmonella cells in DI water		Yang (2008)
	Platinum IDAM were used for the detection of metabolic by-products of the growth of *Listeria innocua* in a low conductivity growth medium	Z_w is Interfacial impedance R_s is bulk solution resistance C_{di} is dielectric capacitance	Gomez et al. (2001)
	Platinum IDAM were used for developing a microscale impedance-based technique for measuring the growth of live cells of *Listeria innocua*, *L. monocytogenes*, and *E. coli* O157:H7 in a low conductivity growth medium		Gomez et al. (2002)
	Gold IDAM were used for the direct detection of *E. coli* K12 by immobilizing anti-*E. coli* antibodies on the surface of glass	R_{sol} is solution resistance C_{sol} is dielectric capacitance of solution R_{cyt} is resistance of cytoplasm R_{blm} is resistance of cell membrane C_{blm} is capacitance of cell membrane C_{par} is parasitic capacitance	Radke and Alocilja (2004)
	Aluminum IDAM were used for the concentration of cells by dielectrophoresis followed by an impedance detection of metabolic by-products of the growth of *Listeria innocua*	C_{di} is dielectric capacitance R_l is bulk resistance R_w is resistance of on-chip wiring Z_1, Z_2 are interfacial impedances	Gomez-Sjöberg et al. (2005)
	Indium tin-oxide IDAM were used for the detection of viable *Salmonella typhimurium* in a non-selective medium after separation with the help of anti-*Salmonella* coated antibodies on the surface of magnetic microbeads	C_{dl} is double layer capacitance R_{sol} is resistance of the medium C_{dl} is dielectric capacitance of medium	Yang and Li (2006)
	Indium tin-oxide IDAM were used for the detection of viable *Salmonella typhimurium* in a selective medium		Yang et al. (2004b)
	Gold IDAM were used for the direct detection of *E. coli* O157:H7 using heterobifunctional cross-linkers and immobilized anti-*E. coli* antibodies on the surface of glass		Radke and Alocilja (2005a)

Table 2 (*Continued*)

Equivalent circuits	Typical applications	Electrical circuit elements	References
	Microfluidic flow cell with embedded gold IDAM were used for the detection of *E. coli* O157:H7. Magnetic nanoparticles-antibody conjugates were used for the separation and concentration of bacterial cells before injecting them into the flow cell		Varshney et al. (2007b)
	Indium tin-oxide IDAM were used for the detection of *E. coli* O157:H7 using epoxisilane and anti-*E. coli* antibodies on the surface of electrodes and employing a redox probe	C_{dl} is double layer capacitance R_s is the resistance of the solution Z_w is Warburg impedance R_{et} is electron transfer resistance	Yang and Li (2005)
	Gold IDAM chip was used for the detection of *E. coli* O157:H7. Magnetic nanoparticles-antibody conjugates were used for the separation and concentration of bacterial cells before injecting them into the flow cell	C_{dl} is double layer capacitance R_{sol} is resistance of the medium C_s is stray capacitance of system	Varshney and Li (2007a)

or finding novel separation techniques will aid in the development of an impedance biosensor with higher sensitivity and lower detection limits.

7. Upcoming trends

Impedance detection is an ever developing field where newer techniques are being introduced while at the same time functionalities of earlier techniques are being improved. In order to achieve lower detection limits, shorter detection times, and improved functionalities, several newer fields can be combined with impedance detection technique. New fields are aiming to find better substitutes for antibodies and nucleic acid probes, improve sampling methods, design low conductivity media, and develop novel nanomaterials (nanotubes, nanofibers, and nanowires) for biomolecular detection.

Bacteriophages (phage) are obligate intracellular parasites of bacteria that multiply by using some or all of the host bio-synthetic machinery. There are two steps involved in the action of bacteriophages on bacteria—selective binding to the bacterial cell and then multiplication of virulent phages and lysis of the bacterial cell. The host specificity of the bacteriophage is against outer membrane of the cell-surface proteins, including lipopolysaccharides, pili, and lipoproteins. Phages can replace antibodies in terms of specificity for the bacterial cells with distinctive additional features of phage multiplication and lysis of host cell. They have additional advantages over antibodies in terms of stability, standardization, and low-cost production (Neufeld et al., 2003; Olsen et al., 2005; Sorokulova et al., 2005; Kretzer et al., 2007). They are more thermostable than antibodies and can function even after exposure to high temperatures during shipping, handling, and storage (Brigati and Petrenko, 2005). Lysis of the cell releases highly conductive cytoplasm and other intracellular components of bacterial cells into the outer medium, which can increase conductivity of the medium as compared to conductivity of the medium with intact cells. Thus phages can potentially add one more way to amplify the impedance signal. The combination of phage-specific identification and the release of intrinsic substances following lysis of the cell provide a powerful tool for highly specific and sensitive detection of a given bacterial strain (Neufeld et al., 2003). Since phages will only replicate in metabolically active bacterial cells, viability can be tested by detecting lytic products produced by phages during their replication cycle. Enzyme labels against different cell markers of bacterial cells can also be used to amplify impedance signal.

Aptamers can also replace antibodies and could prove to be useful in biosensor applications. These artificial nucleic acid ligands can be generated against amino acids, drugs, proteins and other molecules, and thus can be applied for the detection of various molecular targets such as small molecules, proteins, nucleic acids, and even cells, tissues, and organisms (O'Sullivan, 2002; Ngundi et al., 2006). In addition to their discriminate recognition, aptamers offer advantages over antibodies as they can be engineered completely in a test tube, are readily produced by chemical synthesis, and possess desirable storage properties. They can be used for the detection of small molecules by sandwich-type formats, applied for the selection of target analytes in real matrix, modified during immobilization without affecting affinity, and subjected to repeated cycles of denaturation and regeneration.

Nanotechnology has been combined with microfabrication for the development of high-density biosensor arrays composed of nanotubes, nanowires, or nanofibers, each modified with distinct biomolecular recognition elements (Cui et al., 2001; Koehne et al., 2004; Kim et al., 2006; Mauyama et al., 2006; Sharma et al., 2006; Su et al., 2006). Carbon nanotubes and nanofibers are especially attractive for nanoscale biosensing systems because of their exceptional chemical and biochemical stability (Kong and Franklin, 2000; Williams et al., 2002; Poh et al., 2004). The technologies are quite mature for (1) growing single-walled carbon nanotubes, (2) vertically aligning carbon nanofibers on surfaces, and (3) surface modification in order to capture target analyte (Lee et al., 2004). The surface of nanomaterials can be modified to combine with DNA, proteins, and antibodies, and thus find a variety of applications for the detection of DNA, bacterial cells, toxic gases, and a range of biomolecules (Shim et al., 2002; Qi et al., 2003; Kim et al., 2006). Additionally, integrating nanofibers/nanotubes with microelectrodes will result in nanoscale electrodes which do not require difficult nanofabrication techniques.

8. Conclusions

Impedance detection technique attracts much interest from researchers in academic institutions, industries, and government agencies due to its simplicity in design, impressive sensitivity and detection limit. Electrochemical biosensors have been studied for many years at a developmental and research level, successfully applied in industries in past few years, and now accepted as a standard method for screening some bacterial cells in food samples. On one hand, IDAM based impedance techniques have been suitably applied in conventional electrochemical cell formats without the need of flow-based systems, while on the other hand, as the cost of microfluidic devices has been reduced due to advanced microfabrication technology, IDAM techniques can be applied to the development of total analysis systems which employ serial processing of the sample from sampling, purification, and concentration to detection, as well the parallel processing to handle multiple samples. Impedance techniques are also suitable for label-free detection and have important advantages such as speed, de-skilled analysis, fewer numbers of steps, and the potential for the multi-analyte detection.

The large amount of recent activity in impedance sensor research reflects the importance of the field and as well as the power and convenience that IDAM-based impedance detection offers in screening of bacterial cells. There are many quality papers published in the basic research and applied fields to support and advance this technology to the next level and find more applications. However, it is advisable for researchers to put efforts together to apply this technique for real-world applications that present clear challenges on selectivity, sensitivity, detection limit, and ruggedness.

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